

Problemas encontrados y soluciones aplicadas

dealerADMIN - E2E general completamente local 12 de septiembre de 2026

RESULTADO	PASS AC-01 a AC-20
E2E	34 casos PASS 0 FAIL 0 BLOCKED
REGRESION	333 tests PASS en 28 archivos

1. Resumen ejecutivo

Se levantaron PostgreSQL Docker, API y frontend locales. Se enviaron Customer Replied firmados a las fuentes configuradas y se probaron Messenger, WhatsApp, respuestas contaminantes y validas, reparacion por poll, ventanas de 15 segundos/30 segundos/30 minutos/3 horas, concurrencia, duplicados, ruteo y la regla nueva de telefono enviado hace al menos tres dias.

Se encontraron ocho problemas durante la ejecucion. Cada uno fue capturado antes de corregirse, se aplico una correccion acotada y se reejecuto el caso relacionado. La reejecucion final fue 34/34 E2E PASS y la regresion completa fue 333/333 tests PASS. No quedaron pendientes reales ni bloqueos.

La especificacion PDF adjunta se trato como plan de QA. No se activaron GHSL, Neon ni cuentas reales.

2. Version exacta probada

Elemento	Valor
Commit	3470776ed3ec0e72adb4583fa015291a9e9e3985
Docker	c20d4b05ae50 dealeradmin-postgres postgres:18 activo
API / frontend	http://127.0.0.1:3010 / http://127.0.0.1:3000
Migracion	RepairCustomerRepliedSourceAliases1710000020000
Configuracion	NODE_ENV development; America/Bogota; reloj fijo de QA; HMAC simulado no impreso

Esquema verificado: conversation_messages, conversations, dealer_location_aliases, dealer_round_robin_state, dealers, lead_dealers, locations, leads, migrations y webhook_events. Namespace QA final: 37 leads, 45 eventos y 34 conversaciones.

3. Matriz AC-01 a AC-20

Timestamp de evidencia: 2026-09-12T14:06:47.690Z. Los hashes completos, IDs, snapshots y SQL estan en evidence.json.

AC	Resultado	Timestamp	Evidencia
AC-01	PASS	2026-09-12T14:06:47.690Z	Trazabilidad completa
AC-02	PASS	2026-09-12T14:06:47.690Z	HMAC invalido
AC-03	PASS	2026-09-12T14:06:47.690Z	Idempotencia exacta
AC-04	PASS	2026-09-12T14:06:47.690Z	Identidad Messenger/WhatsApp
AC-05	PASS	2026-09-12T14:06:47.690Z	No contaminar real_name
AC-06	PASS	2026-09-12T14:06:47.690Z	Evidencia de vehiculo
AC-07	PASS	2026-09-12T14:06:47.690Z	Down y timeline separados
AC-08	PASS	2026-09-12T14:06:47.690Z	Documentos no bloquean
AC-09	PASS	2026-09-12T14:06:47.690Z	Reparacion poll 30 s
AC-10	PASS	2026-09-12T14:06:47.690Z	No mezclar campos
AC-11	PASS	2026-09-12T14:06:47.690Z	Ventanas Bogota
AC-12	PASS	2026-09-12T14:06:47.690Z	waiting_window null
AC-13	PASS	2026-09-12T14:06:47.690Z	Action Cars por idioma
AC-14	PASS	2026-09-12T14:06:47.690Z	Millersville alternancia
AC-15	PASS	2026-09-12T14:06:47.690Z	Easterns ciudad/estado
AC-16	PASS	2026-09-12T14:06:47.690Z	Duplicado queued
AC-17	PASS	2026-09-12T14:06:47.690Z	Coherencia de estado
AC-18	PASS	2026-09-12T14:06:47.690Z	Sin fuga dealer
AC-19	PASS	2026-09-12T14:06:47.690Z	Sent y concurrencia
AC-20	PASS	2026-09-12T14:06:47.690Z	Frontend refleja BD

4. Detalle de cada conversacion

Cada fila resume payload hash, event_id, conversation_id, estados, dealer, ventana y routing_reason. Los message IDs completos, snapshots before/after, location_snapshot y queries estan en evidence.json.

Case	Conversation	Payload/event	Estado	Dealer/ventana	Resultado
qa-e2e-20260912-source-stafford	qa-e2e-20260912-source-stafford-conversation	event: qa-e2e-20260912-source-stafford-evt-01, qa-e2e-20260912-source-stafford-evt-02, qa-e2e-20260912-source-stafford-evt-03, qa-e2e-20260912-source-stafford-evt-04, qa-e2e-20260912-source-stafford-evt-05, qa-e2e-20260912-source-stafford-evt-06 hash: 32ad631dfe4e628a, 2fc72e8544f72d80, 3255cea600ec9318, 65f9e32958729a1e, 0a16a85171c4c189, c2c3818727d0ae81	before=None after=queued	dealer=['3bf48365-6833-4fb7-b401-1d6ea04ba975'] next=None ['GHL stafford source dealer']	source stafford persisted and queued
qa-e2e-20260912-source-fredericksburg	qa-e2e-20260912-source-fredericksburg-conversation	event: qa-e2e-20260912-source-fredericksburg-evt-01 hash: 91c8ecbda4da4b8f	before=None after=queued	dealer=['dc621cd1-a389-478-5-b7ee-20cc2c7a0a15'] next=None ['GHL fredericksburg source dealer']	source fredericksburg persisted and queued
qa-e2e-20260912-source-fredericksburg_2	qa-e2e-20260912-source-fredericksburg_2-conversation	event: qa-e2e-20260912-source-fredericksburg_2-evt-01 hash: 1157448bc1442824	before=None after=queued	dealer=['5445801a-4aa8-49f0-868d-80479a823f78'] next=None ['GHL fredericksburg-2 source dealer']	source fredericksburg-2 persisted and queued
qa-e2e-20260912-source-arlington	qa-e2e-20260912-source-arlington-conversation	event: qa-e2e-20260912-source-arlington-evt-01 hash: c6950b94b68c25d8	before=None after=queued	dealer=['d4444444-4444-4444-4444-444444444444'] next=None ['GHL arlington source dealer']	source arlington persisted and queued
qa-e2e-20260912-source-koons_fred	qa-e2e-20260912-source-koons_fred-conversation	event: qa-e2e-20260912-source-koons_fred-evt-01 hash: 56425c695842f9df	before=None after=queued	dealer=['e5555555-5555-5555-5555-555555555555'] next=None ['GHL koons-fred source dealer']	source koons-fred persisted and queued
qa-e2e-20260912-source-koons_fred_eng	qa-e2e-20260912-source-koons_fred_eng-conversation	event: qa-e2e-20260912-source-koons_fred_eng-evt-01 hash: c2051849a1fdf303	before=None after=queued	dealer=['e6666666-6666-6666-6666-666666666666'] next=None ['GHL koons-fred-eng source dealer']	source koons-fred-eng persisted and queued
qa-e2e-20260912-source-koons_culpeper	qa-e2e-20260912-source-koons_culpeper-conversation	event: qa-e2e-20260912-source-koons_culpeper-evt-01 hash: e2237e74b114eab1	before=None after=queued	dealer=['e7777777-7777-7777-7777-777777777777'] next=None ['GHL koons-culpeper source dealer']	source koons-culpeper persisted and queued
qa-e2e-20260912-source-easterns_frederick	qa-e2e-20260912-source-easterns_frederick-conversation	event: qa-e2e-20260912-source-easterns_frederick-evt-01 hash: 4ffe246f1ae994e2	before=None after=queued	dealer=['eccccccc-cccc-cccc-cccc-cccccccccccc'] next=None ['GHL easterns-frederick source dealer']	source easterns-frederick persisted and queued
qa-e2e-20260912-action-es	qa-e2e-20260912-action-es-conversation	event: qa-e2e-20260912-action-es-evt-01 hash: 09d731a613873dcf	before=None after=queued	dealer=['e8888888-8888-8888-8888-888888888888'] next=None ['GHL action-cars source dealer']	Action Cars Spanish queue
qa-e2e-20260912-action-en	qa-e2e-20260912-action-en-conversation	event: qa-e2e-20260912-action-en-evt-01 hash: 93e2037e34af2ab1	before=None after=queued	dealer=['e9999999-9999-9999-9999-999999999999'] next=None ['GHL action-cars source dealer']	Action Cars English queue
qa-e2e-20260912-millersville-1	qa-e2e-20260912-millersville-1-conversation	event: qa-e2e-20260912-millersville-1-evt-01 hash: 61dfab5b4a49d00a	before=None after=queued	dealer=['eaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa'] next=None ['GHL easterns-millersville source dealer']	Millersville case 1 queued
qa-e2e-20260912-millersville-2	qa-e2e-20260912-millersville-2-conversation	event: qa-e2e-20260912-millersville-2-evt-01 hash: d38759d966b307fd	before=None after=queued	dealer=['ebbbbbbb-bbbb-bbbb-b-bbbb-bbbbbbbbbbbb'] next=None ['GHL easterns-millersville source dealer']	Millersville case 2 queued
qa-e2e-20260912-millersville-3	qa-e2e-20260912-millersville-3-conversation	event: qa-e2e-20260912-millersville-3-evt-01 hash: a41352c500032fe4	before=None after=queued	dealer=['eaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa'] next=None ['GHL easterns-millersville source dealer']	Millersville case 3 queued

Case	Conversation	Payload/event	Estado	Dealer/ventana	Resultado
qa-e2e-20260912-millerville-4	qa-e2e-20260912-millersville-4-conversation	event: qa-e2e-20260912-millersville-4-evt-01 hash: 487434bbffa808f5	before=None after=queued	dealer=['ebbbbbbb-bbbb-bbb b-bbbb-bbbbbbbbbb'] next=None ['GHL easterns-millersville source dealer']	Millersville case 4 queued
qa-e2e-20260912-easterns-rosedale	qa-e2e-20260912-easterns-rosedale-conversation	event: qa-e2e-20260912-easterns-rosedale-evt-01 hash: 7edd1b7bb19ef1a0	before=None after=queued	dealer=['d1111111-1111-1111-1111-1111111111'] next=None ['Source dealer from GHL location (fallback)']	Easterns rosedale expected dealer d11111 11-1111-1111- 1111-1111111 11111
qa-e2e-20260912-easterns-laurel-ambiguous	qa-e2e-20260912-easterns-laurel-ambiguous-conversation	event: qa-e2e-20260912-easterns-laurel-ambiguous-evt-01 hash: b9d33fc604af7184	before=None after=queued	dealer=['d2222222-2222-2222-2222-222222222222'] next=None ['Exclusive Zone: Laurel, Maryland']	Easterns laurel -ambiguous expected dealer d22222 22-2222-2222- 2222-2222222 22222
qa-e2e-20260912-easterns-laurel-explicit-va	qa-e2e-20260912-easterns-laurel-explicit-va-conversation	event: qa-e2e-20260912-easterns-laurel-explicit-va-evt-01 hash: bbeb972cee3fb531	before=None after=queued	dealer=['d3333333-3333-3333-3333-3333333333333333'] next=None ['Exclusive Zone: State Virginia']	Easterns laurel -explicit-va expected dealer d33333 33-3333-3333- 3333-3333333 33333
qa-e2e-20260912-easterns-sterling	qa-e2e-20260912-easterns-sterling-conversation	event: qa-e2e-20260912-easterns-sterling-evt-01 hash: 1eea6f2692342813	before=None after=queued	dealer=['d3333333-3333-3333-3333-333333333333'] next=None ['Exclusive Zone: State Virginia']	Easterns sterling expected dealer d33333 33-3333-3333- 3333-3333333 33333
qa-e2e-20260912-easterns-newark-collision	qa-e2e-20260912-easterns-newark-collision-conversation	event: qa-e2e-20260912-easterns-newark-collision-evt-01 hash: 73adcbd46e5839f	before=None after=queued	dealer=['d1111111-1111-1111-1111-111111111111'] next=None ['Source dealer from GHL location (fallback)']	Easterns newa rk-collision expected dealer d11111 11-1111-1111- 1111-1111111 11111
qa-e2e-20260912-late-repair	qa-e2e-20260912-late-repair-conversation	event: qa-e2e-20260912-late-repair-evt-01, qa-e2e-20260912-late-repair-evt-02 hash: c3045f34047671bf, 5f5060e45c388f5d	before= after=queued	dealer=[] next= []	late inbound message repaired same conversation after simulated 30s poll
qa-e2e-20260912-manual-waiting-null	qa-e2e-20260912-manual-waiting-null-conversation	event: qa-e2e-20260912-manual-waiting-null-evt-01 hash: 09bccd7f748a254a	before=None after=waiting_window	dealer=[] next=2026-09-12T15:30:00.00Z []	manual waiting _window null received calculated next _attempt_at
qa-e2e-20260912-window-day	qa-e2e-20260912-window-day-conversation	event: qa-e2e-20260912-window-day-evt-01 hash: 262b8adc1472b352	before=None after=queued	dealer=[] next=2026-09-12T15:00:15.00Z []	10:00 Bogota plus simulated 30 minutes queued
qa-e2e-20260912-window-night	qa-e2e-20260912-window-night-conversation	event: qa-e2e-20260912-window-night-evt-01 hash: ce4c9d497ace2019	before=None after=queued	dealer=[] next=2026-09-13T01:00:15.00Z []	20:00 Bogota plus simulated 3 hours queued
qa-e2e-20260912-contamination-question	qa-e2e-20260912-contamination-question-conversation	event: qa-e2e-20260912-contamination-question-evt-01, qa-e2e-20260912-contamination-question-evt-02 hash: 038ed96b78c11b24, 8b2836560f612ae7	before=None after=partial	dealer=[] next=None []	bot question and financing intent stay out of name/vehicle
qa-e2e-20260912-isolated-mustang	qa-e2e-20260912-isolated-mustang-conversation	event: qa-e2e-20260912-isolated-mustang-evt-01 hash: 54af501c84ece002	before=None after=waiting_window	dealer=[] next=2026-09-12T15:00:15.00Z []	isolated Mustang is not vehicle evidence
qa-e2e-20260912-years-and-down	qa-e2e-20260912-years-and-down-conversation	event: qa-e2e-20260912-years-and-down-evt-01, qa-e2e-20260912-years-and-down-evt-02, qa-e2e-20260912-years-and-down-evt-03 hash: 125dce7866b1a856, 4c857c385e432ec2, 11380d42dcf1161c	before=None after=waiting_window	dealer=[] next=2026-09-12T15:00:15.00Z []	years not down and 2000 is down

Case	Conversation	Payload/event	Estado	Dealer/ventana	Resultado
qa-e2e-20260912-phone-correction	qa-e2e-20260912-phone-correction-conversation	event: qa-e2e-20260912-phone-correction-evt-01, qa-e2e-20260912-phone-correction-evt-02, qa-e2e-20260912-phone-correction-evt-03 hash: 2ed3dd3ac9f863d8, 274241a6438a49b7, 00d331c2bee7a824	before= after=waiting_window	dealer=[] next=[]	invalid phone corrected by latest valid inbound number
qa-e2e-20260912-duplicate-original	qa-e2e-20260912-duplicate-original-conversation	event: qa-e2e-20260912-duplicate-original-evt-01 hash: a8b923d594702200	before=None after=queued	dealer=['dc621cd1-a389-4785-b7ee-20cc2c7a0a15'] next=None ['GHL fredericksburg source dealer']	duplicate source queued first
qa-e2e-20260912-duplicate-retry	qa-e2e-20260912-duplicate-retry-conversation	event: qa-e2e-20260912-duplicate-retry-evt-01 hash: d6c0bcacb6985d60	before=None after=duplicate_ignored	dealer=[] next=None []	queued same name and phone ignored
qa-e2e-20260912-idempotent-message-retry	qa-e2e-20260912-idempotent-message-retry-conversation	event: qa-e2e-20260912-idempotent-message-retry-evt-01, qa-e2e-20260912-idempotent-message-retry-evt-01 hash: bc85f0fd56f5c0a4, bc85f0fd56f5c0a4	before= after=waiting_window	dealer=[] next=[]	same event and message retry is idempotent
qa-e2e-20260912-sent-reprocess	qa-e2e-20260912-sent-reprocess-conversation	event: qa-e2e-20260912-sent-reprocess-evt-01 hash: 1b8cc55f49b99b20	before= after=sent	dealer=[] next=[]	sent reprocess response {"accepted":true,"eventId":"qa-e2e-20260912-sent-reprocess-evt-01","conversationId":"00000000-0-0000-0000-0000-000000000004","source":"fredericksburg","status":"processed"}
qa-e2e-20260912-concurrent-poll	qa-e2e-20260912-concurrent-poll-conversation	event: qa-e2e-20260912-concurrent-poll-evt-01 hash: 3b4b27645bf14fdf	before= after=queued	dealer=[] next=[]	concurrent poll produces one queued relationship
qa-e2e-20260912-stale-phone-exact-3-days	qa-e2e-20260912-stale-phone-exact-3-days-conversation	event: qa-e2e-20260912-stale-phone-exact-3-days-evt-01 hash: 4b0694ee302ee3e7	before= after=stale_phone_ignored	dealer=[] next=[]	phone sent exactly 3 days earlier is not accepted as new lead
qa-e2e-20260912-recent-phone-allowed	qa-e2e-20260912-recent-phone-allowed-conversation	event: qa-e2e-20260912-recent-phone-allowed-evt-01 hash: 6f0e54e37c906b68	before= after=waiting_window	dealer=[] next=[]	phone sent less than 3 days earlier is not blocked by stale rule

5. Incidentes de normalizacion

Se aplico la regla conservadora: si no hay evidencia suficiente, el campo queda vacio. No se mezclan real_name, vehicle_type, down_payment, purchase_timeline, documents, identification ni bank_account.

Se verificaron nombres declarados, vehiculos Mustang/truck/SUV/Sedan, down payments 1000/2000/3000, anios 2025/2018, plazos, identificacion, cuenta bancaria, documentos y prueba de ingresos. Tambien se corrigio telefono invalido seguido de telefono valido conservando la misma conversacion.

Regla nueva: sent_at de al menos tres dias produce stale_phone_ignored y no crea nueva cola; sent_at de dos dias permite el lead. Casos: stale-phone-exact-3-days y recent-phone-allowed.

6. Action Cars y Millersville

Action Cars uso el mismo endpoint con action-es y action-en; cada idioma quedo en su dealer/cola sin mezcla.

Millersville produjo exactamente: Easterns Millersville, White Marsh, Millersville, White Marsh.

7. Ruteo Easterns

Rosedale fue a Rosedale; Laurel ambiguo priorizo Maryland y fue a Laurel; Laurel VA respetó el estado explícito y fue a Sterling; Sterling fue a Sterling; Newark en colisión respetó la prioridad MD. Un nombre de contacto que contenia Easterns Laurel no selecciono dealer.

8. Pruebas de 30 segundos y ventanas

Prueba de 30 segundos: se dejo una conversacion incompleta, se agrego el inbound tardio 2000/esta semana y process-due a +30 s reparo la misma conversation_db_id sin duplicarla.

Ventanas distintas: 10:00 America/Bogota +30 min y 20:00 America/Bogota +3 h. La estabilizacion de +15 s y el poll de +30 s se probaron por separado.

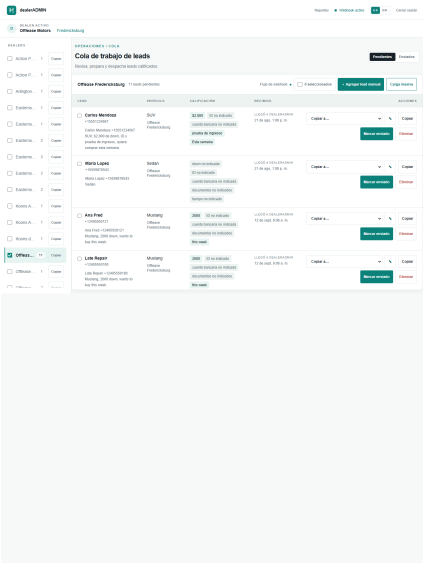
Tambien se cambio manualmente waiting_window con next_attempt_at null y el proceso calculo la fecha con reloj controlado.

9. Regresiones y pendientes reales

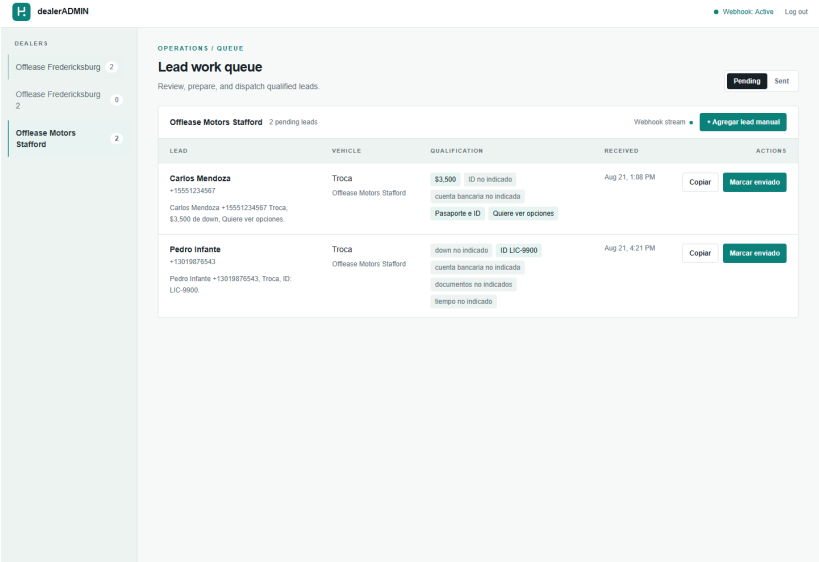
Despues de cada correccion se reejecuto el caso fallido y regresiones relacionadas. Resultado: 34/34 E2E PASS y 333/333 tests PASS, incluyendo carga de 150 webhooks, duplicados, orden invertido y rollback. Pendientes reales: ninguno para AC-01 a AC-20.

10. Capturas incluidas

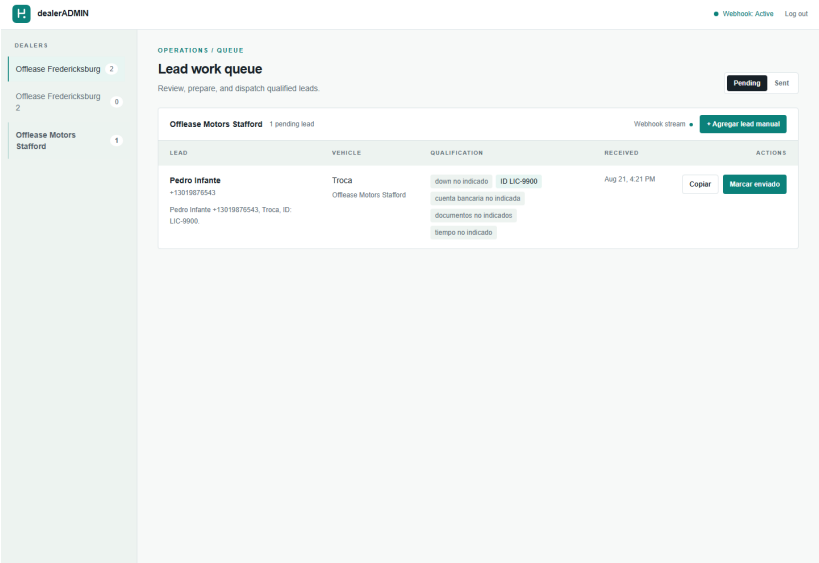
Capturas del estado local de UI y BD que acompañan este informe:



Frontend local autenticado: cola Fredericksburg, Ana Fred y Late Repair



BD local antes del webhook



BD local despues del webhook, pending

dealerADMIN

Webhook: Active Log out

DEALERS

Offlease Fredericksburg 2

Offlease Fredericksburg 2

Offlease Motors Stafford 1

OPERATIONS / QUEUE

Lead work queue

Review, prepare, and dispatch qualified leads.

Pending Sent

Offlease Motors Stafford 1 sent lead

Webhook stream Agregar lead manual

LEAD	VEHICLE	QUALIFICATION	RECEIVED	ACTIONS
Carlos Mendoza +15551234567 Carlos Mendoza +15551234567 Troca, \$3,500 de down. Quiera ver opciones.	Troca Offlease Motors Stafford	\$3,500 ID no indicado cuenta bancaria no indicada Pasaporte e ID Quiera ver opciones	Aug 21, 1:08 PM	Copiar

BD local despues del reprocesamiento, sent

11. Detalle de incidentes y soluciones

F-001 - migration

Observado: Migration GhLocationRouting1710000007000 failed because relation dealer_location_aliases did not exist.

Causa raíz: Local database had migration 0003 recorded as applied while its prerequisite table was absent.

Correccion: Reassert dealer_location_aliases with CREATE TABLE IF NOT EXISTS before migration 0007.

Archivos: apps/api/src/database/migrations/1710000007000-GhLocationRouting.ts

Evidencia/SQL: SELECT table_name FROM information_schema.tables WHERE table_schema='public' AND table_name IN ('dealer_location_aliases','locations');

Retest: Migrations completed; final migration RepairCustomerRepliedSourceAliases1710000020000 is applied.

F-002 - source-probe

Observado: Customer Replied for source fredericksburg returned HTTP 400: no active dealer for MyxWNKacThim798E8KC6.

Causa raíz: The two Fredericksburg source aliases were missing despite the historical migration state.

Correccion: Added an idempotent repair migration for both source aliases and registered it in AppModule and AppDataSource.

Archivos: apps/api/src/database/migrations/1710000020000-RepairCustomerRepliedSourceAliases.ts, apps/api/src/app.module.ts, apps/api/src/database/data-source.ts

Evidencia/SQL: SELECT gh_location_id,dealer_id FROM dealer_location_aliases WHERE gh_location_id IN ('MyxWNKacThim798E8KC6','bAuMEQeH48xAtu9tAMFF');

Retest: All configured sources accepted and queued in the final 34-case run.

F-003 - late-repair

Observado: The second inbound message '2000\nesta semana' left down_payment empty and the conversation stayed waiting_window.

Causa raíz: Webhook persistence applied whitespace flattening before transcript reconstruction, destroying standalone numeric message boundaries.

Correccion: Preserved normalized line breaks in conversation_messages and transcript assembly.

Archivos: apps/api/src/features/webhooks/application/conversation-webhook.service.ts, apps/api/src/features/leads/tests/unit/collector-normalizer.spec.ts

Evidencia/SQL: SELECT gh_message_id,replace(body,E'\n','<NL>') AS body,qualification_snapshot FROM conversation_messages JOIN conversations ON conversations.id=conversation_messages.conversation_id WHERE gh_conversation_id='qa-e2e-20260912-late-repair-conversation';

Retest: Late repair became queued on the same conversation after simulated 30s; unit regression passes.

F-004 - Easterns explicit state

Observado: Laurel VA was assigned to Laurel with routing_reason 'Explicit Easterns Zone: Laurel → Laurel'.

Causa raíz: A broad qualification-memory regex treated contact name 'Easterns Laurel-Explicit-Va' as explicit dealer selection.

Correccion: Require an explicit conversational selection phrase; contact names cannot select a dealer.

Archivos: apps/api/src/features/routing/domain/services/georouting.service.ts, apps/api/src/features/routing/tests/unit/georouting.spec.ts

Evidencia/SQL: SELECT location_snapshot,qualification_snapshot->>'assigned_dealer_id',assigned_dealer_id,routing_reason FROM conversations JOIN leads ON leads.id=conversations.lead_id JOIN lead_dealers ON lead_dealers.lead_id=leads.id WHERE gh_conversation_id='qa-e2e-20260912-easterns-laurel-explicit-va-conversation';

Retest: Laurel VA assigned Sterling in the final run; regression test passes.

F-005 - Easterns ambiguous location

Observado: Laurel without state fell back to source Rosedale instead of the MD-priority Laurel queue.

Causa raíz: Georouting resolved MD but had no Laurel-MD branch, so it reached the source-anchor fallback.

Correccion: Added the Laurel/MD geographic branch after explicit-state handling.

Archivos: apps/api/src/features/routing/domain/services/georouting.service.ts, apps/api/src/features/routing/tests/unit/georouting.spec.ts

Evidencia/SQL: SELECT location_snapshot,assigned_dealer_id,routing_reason FROM conversations JOIN leads ON leads.id=conversations.lead_id JOIN lead_dealers ON lead_dealers.lead_id=leads.id WHERE ghl_conversation_id='qa-e2e-20260912-easterns-laurel-ambiguous-conversation';

Retest: Ambiguous Laurel assigned Laurel; final Easterns cases all pass.

F-006 - unit regression

Observado: One unit test still expected CURRENT_TIMESTAMP after controlled-clock logic was introduced.

Causa raiz: Test assertion lagged the intentional parameterized timestamp contract.

Correccion: Updated the assertion to require SET next_attempt_at = \$1::timestampz and the fixed parameter.

Archivos: apps/api/src/features/webhooks/tests/unit/conversation-webhook.service.spec.ts

Evidencia/SQL: evidence.json y logs locales

Retest: Full suite with resilience enabled: 333 passed across 28 test files.

F-007 - frontend local process

Observado: The first authenticated UI attempt saw 404 on /api/auth/login and later a selector timeout.

Causa raiz: Frontend production build was created without API_URL rewrite; one temporary QA selector matched a hidden mobile duplicate.

Correccion: Rebuilt web with API_URL=http://127.0.0.1:3010, restarted Next, and made the selector target the visible table.

Archivos: apps/web/next.config.js, tmp/qa-e2e-local-20260912/check-frontend.mjs

Evidencia/SQL: evidence.json y logs locales

Retest: Frontend login/dashboard PASS; local lead rendered and screenshot saved.

F-008 - resilience integration

Observado: The out-of-order qualification test expected raw '\$4,500', 'ID-OUT-OF-ORDER', 'Cuenta verificada' and 'Este mes', but the normalized contract returned '4500', 'yes', '', and 'this month'.

Causa raiz: The existing stress-test assertion was written for pre-normalization values; the service intentionally stores canonical qualification values and does not infer an unrecognized bank-account phrase.

Correccion: Aligned the test expectation and message assertion with the normalized storage contract.

Archivos: apps/api/src/features/leads/tests/integration/resilience-stress.spec.ts

Evidencia/SQL: evidence.json y logs locales

Retest: The four resilience tests pass against the local PostgreSQL container.

12. Cierre

La implementacion queda validada localmente con PASS AC-01 a AC-20. El PDF no contiene secretos; los datos y teléfonos mostrados son fixtures del namespace QA.

CIERRE

PASS | 0 FAIL | 0 BLOCKED